

Video Transcripts

Week 2.2: The Full Generative AI Stack – Enabling & Scaling Fundamentals

Video 1: The Layered GenAI Stack

Hello learners, and welcome to this week. When a developer uses GitHub Copilot and receives a code suggestion instantly, the experience appears simple. In reality, that moment depends on a layered architecture known as the generative AI stack. These systems operate through multiple layers that transform raw computation into usable intelligence. A practical way to understand this architecture is through seven layers: hardware, GPU drivers, machine learning frameworks, foundation models, adaptation techniques such as retrieval-augmented generation or fine-tuning, inference and serving systems, and finally the applications that users interact with.

The stack begins with hardware. Large language models rely on massive matrix computations across billions of parameters, which require highly parallel processing. Standard CPUs struggle with this, which is why GPUs have become central to modern AI infrastructure. Companies such as NVIDIA produce specialised chips designed for deep learning, including the NVIDIA A100 and the NVIDIA H100. The A100 delivers roughly 312 teraflops of AI compute, while the H100 can exceed 1,000 teraflops for transformer workloads, though it consumes significantly more power. Training frontier-scale models comparable to GPT-4 can require tens of thousands of GPUs running continuously for weeks, consuming vast amounts of electricity. As a result, only a small number of organisations, including OpenAI, Google, and Meta, can realistically train models at this scale.

For most organisations, however, the challenge is not training models but selecting the right compute for real workloads. A mid-sized company handling thousands of daily support queries may not require the most advanced GPUs. While high-end infrastructure offers strong performance, factors such as cost, availability, and deployment timelines often lead organisations to choose more practical options. Cloud infrastructure becomes a major expense, with GPU-backed instances on platforms like Microsoft Azure and Google Cloud costing several dollars per hour. At scale, monthly inference costs can quickly reach tens of thousands of dollars, making optimisation of model size and usage critical. This reflects a broader reality: infrastructure decisions are driven as much by cost and accessibility as by raw performance.

Above the hardware layer sits the GPU driver ecosystem, which enables software to interact efficiently with GPU resources. One of the most influential systems here is CUDA, which translates machine learning computations into instructions that GPUs can execute in parallel. Built on top of this are machine learning frameworks such as PyTorch and TensorFlow. These frameworks allow developers to design neural networks, distribute training across multiple GPUs, and optimise large-scale models without needing to manage low-level hardware operations.

The next layer consists of foundation models, which provide the core intelligence behind generative AI systems. Examples include models developed by OpenAI and Google, as well as the LLaMA family released by Meta. Training such models is extremely expensive, often costing tens or even hundreds of millions of dollars, which is why most organisations build on top of existing models rather than developing their own. To make these models useful in specific business contexts, adaptation techniques are applied. Fine-tuning adjusts models using domain-specific data, while retrieval-augmented generation (RAG) allows models to access external knowledge sources. For instance, Klarna has deployed AI assistants that rely on internal documentation to deliver accurate customer support at scale.

Once adapted, models must operate efficiently through the inference and serving layer. These systems manage incoming requests, allocate compute resources, and ensure that responses are generated quickly enough for real-world use. Many enterprise deployments run on cloud platforms such as Microsoft Azure, where large GPU clusters process millions of queries daily. Finally, the application layer is where users interact with generative AI. Tools like GitHub Copilot embed AI directly into everyday workflows, making the technology feel seamless and intuitive. While much of the underlying infrastructure is shared, it is at the application layer that organisations differentiate—turning generative AI into tangible productivity gains and business value.

Understanding the full stack is critical because cost, performance, and reliability depend on decisions made across every layer. Generative AI systems succeed not by optimising a single component, but by carefully designing how the entire stack works together.

Video 2: Where Constraints Originate in the Stack

Imagine rolling out a generative AI system for customer support. Initially, everything appears to improve—queries are handled faster, resolution times drop, and costs seem to decline. However, as usage scales, new challenges emerge: operational costs increase, response times slow during peak demand, and poorly designed prompts risk exposing sensitive data. These constraints arise across the entire generative AI stack and often compound one another. The experience of Klarna illustrates this clearly. After launching an AI assistant powered by OpenAI in 2024, Klarna handled 2.3 million conversations within a month—around two-thirds of all customer interactions—equivalent to the work of hundreds of agents. Resolution times dropped significantly, repeat queries declined, and the company projected substantial financial gains as the system scaled.

However, these gains revealed deeper constraints, particularly at the infrastructure level. Frontier models rely on advanced hardware such as the NVIDIA H100, which is expensive both to purchase and to run in cloud environments like Microsoft Azure and Google Cloud. At high volumes, where millions of queries involve thousands of tokens each, costs can quickly escalate, shifting expenditure from human labour to cloud infrastructure. Klarna’s leadership acknowledged that an early focus on cost reduction led to quality trade-offs, prompting a shift towards a hybrid model that combines AI efficiency with human oversight.

Latency introduces another layer of constraint, particularly during inference and orchestration. While advanced models such as GPT-4o perform strongly on benchmarks like MMLU, they can be slower to respond compared to smaller models. In practice, simple queries may be resolved instantly, but more complex cases involving multiple systems or rules can take several seconds, especially under heavy load. To manage this, many organisations adopt a tiered approach—routing routine queries to faster, lower-cost models such as GPT-4o mini, while reserving more advanced models for complex scenarios. This strategy reduces both latency and cost while maintaining overall performance.

Governance constraints become critical at the model and application layers, particularly in regulated industries. Financial data, for example, must comply with regulations such as GDPR, requiring strict controls on how information is processed and stored. Klarna addressed this by limiting model access to anonymised data, implementing strong prompt controls, and maintaining detailed logging and audit mechanisms. Without such safeguards, organisations risk data breaches, regulatory penalties, and reputational damage. Past incidents, such as biased AI systems or unintended data exposure in enterprise environments, highlight the importance of robust governance frameworks.

These challenges demonstrate that generative AI is not a plug-and-play solution. Decisions made at one layer—such as model selection—directly affect costs, performance, and risk across the entire

stack. Choosing powerful models like Gemini or LLaMA may improve capability but can also increase expense and operational complexity. As Klarna's experience shows, the most effective approach is often a balanced one: combining AI for scale with human expertise for nuanced decision-making. The key takeaway is clear. Generative AI systems must be designed holistically. Cost, latency, and governance constraints originate at different layers but interact continuously. Optimising a single component in isolation is not enough. Organisations that address these constraints across the full stack are far more likely to achieve scalable, reliable, and responsible AI deployments.

Video 3: GPUs vs AI Accelerators

When organisations deploy generative AI, the underlying hardware often remains invisible, yet it directly determines cost, response speed, energy consumption, and overall competitiveness. The core decision typically lies between general-purpose GPUs, led by NVIDIA, and specialised AI accelerators such as Google's Tensor Processing Units or custom chips developed by large technology firms. This choice shapes how efficiently AI systems operate at scale.

GPUs continue to dominate most enterprise deployments. NVIDIA's chips, including the NVIDIA H100 and the newer NVIDIA Blackwell B200, are designed for the parallel computations required by large language models. By 2026, NVIDIA holds a significant share of the AI hardware market, particularly for training advanced models. These chips are expensive to purchase and operate, whether deployed on-premise or rented through cloud platforms. However, their performance and flexibility make them the default choice for many organisations building or scaling AI capabilities.

A key advantage of NVIDIA's ecosystem lies in its software stack, especially CUDA. CUDA enables developers to efficiently program GPUs for AI workloads and is deeply integrated into leading machine learning frameworks such as PyTorch and TensorFlow. This widespread optimisation creates strong developer preference and practical lock-in, as moving to alternative hardware often requires significant redevelopment effort. As a result, many leading AI systems—from those developed by OpenAI to open-source models like LLaMA—are trained on large clusters of NVIDIA GPUs.

In contrast, specialised accelerators are designed for efficiency in targeted environments. Google's Tensor Processing Units, for example, are optimised specifically for neural network operations and power services such as Search, Translate, and the Gemini ecosystem. These chips can deliver strong performance with lower energy consumption and cost per task within controlled infrastructures, although they are less flexible outside those environments. Similarly, companies like Amazon, Microsoft, and Meta are investing in custom chips to optimise inference—the stage where models generate outputs—because it now accounts for the majority of AI compute in production systems.

The distinction between training and inference is critical for business planning. Training large models requires vast computational resources, limiting it to a small number of organisations. Inference, however, prioritises speed, cost efficiency, and low latency. Many organisations optimise for this stage by simplifying models through techniques such as distillation or reduced-precision computation. In high-performance environments, such as autonomous systems developed by Tesla, even minor delays can be unacceptable, making specialised hardware essential.

Another important consideration is memory bandwidth. Modern AI workloads often spend more time moving data than performing calculations. High-bandwidth memory—reaching several terabytes per second—ensures that processors remain fully utilised, significantly improving overall performance. Without sufficient bandwidth, even the most powerful processors can become inefficient.

For business leaders, hardware decisions are strategic rather than purely technical. They influence operating costs, system responsiveness, sustainability goals, and long-term scalability. General-

purpose GPUs, supported by the CUDA ecosystem, offer flexibility across a wide range of use cases. Specialised accelerators, on the other hand, deliver efficiency for high-volume, narrowly defined workloads. Striking the right balance between these options enables organisations to control costs, improve performance, and strengthen their position in the evolving generative AI landscape.

Video 4: Energy Bottlenecks & AI Sovereignty

When organisations push generative AI to new levels, the real constraint is often not better algorithms or more data, but energy. By 2026, AI data centres alone consume around 30 gigawatts globally—comparable to the electricity usage of a country like the Netherlands. Training a single frontier model, such as successors to Grok 3, can require sustained power of 100 to 130 megawatts. This is equivalent to the energy demand of a mid-sized city running continuously for weeks, supported by dense GPU clusters that also require extensive cooling infrastructure.

In response to these pressures, companies are securing long-term energy sources. In 2024, Microsoft signed a 20-year agreement with Constellation Energy to source power from the Three Mile Island Nuclear Generating Station, which is expected to generate around 835 megawatts upon restart. This move supports Microsoft's goal of becoming carbon negative by 2030, even as its energy consumption has increased significantly due to AI and cloud expansion. At an operational level, energy instability can have immediate business consequences. A failed training run due to a power disruption can result in substantial financial losses, while reliance on backup systems such as diesel generators increases both cost and emissions.

Carbon impact further intensifies the challenge. Data centre electricity consumption could reach nearly 945 terawatt hours annually by 2030, often relying on fossil fuels in rapidly growing markets. Google has maintained carbon neutrality since 2007 and is investing heavily in renewable energy to achieve round-the-clock clean power. However, organisations operating in carbon-intensive regions face rising costs due to stricter environmental regulations and carbon pricing mechanisms. For instance, financial institutions implementing AI-driven analytics have already experienced increased costs linked to emissions reporting and sustainability compliance.

Energy considerations are now closely tied to AI sovereignty, as nations seek to control their technological infrastructure. The CHIPS and Science Act allocates substantial funding to domestic semiconductor manufacturing, while the European Union's AI Act combines regulatory oversight with efforts to strengthen local capabilities. In India, the Semiconductor Mission is driving investment in local chip production, even as data centre capacity is expected to expand rapidly. However, grid limitations and power disruptions in major cities such as Mumbai and Delhi can delay AI deployments, prompting companies like Reliance Industries to invest in large-scale renewable energy projects to support future demand.

Globally, countries such as Saudi Arabia are also investing heavily in AI infrastructure through initiatives led by the Public Investment Fund. Projects like NEOM aim to build technology ecosystems powered largely by renewable energy, creating opportunities for regional AI development, including language models tailored to local markets. At the same time, vendor concentration adds another layer of risk, with companies like NVIDIA dominating high-performance AI hardware and cloud providers such as Amazon Web Services, Microsoft Azure, and Google Cloud controlling a large share of workloads. Geopolitical factors, including export restrictions, can further constrain access to critical infrastructure.

For business leaders, the message is clear: generative AI success depends as much on energy strategy as on technical capability. Organisations must plan for renewable energy integration, diversify

infrastructure and suppliers, and incorporate sovereignty considerations into long-term strategy. Ignoring these factors risks operational disruption and rising costs, while addressing them enables resilient, scalable, and sustainable AI growth.

Video 5: Cloud & OEM Infrastructure: Hyperscalers as the Scaling Layer

When organisations build generative AI systems, the real challenge begins after the model is trained. Turning a prototype into a reliable, global service requires massive scale, resilience, and continuous optimisation. This is where hyperscalers such as Microsoft Azure, Amazon Web Services, and Google Cloud Platform play a critical role. They provide the scaling layer—handling training, fine-tuning, deployment, and monitoring—without requiring organisations to build and maintain their own data centres.

A strong real-world example is GitHub Copilot. By early 2026, it serves tens of millions of users, with developers relying on it daily for code suggestions. The system must respond in under a second, even during peak usage across multiple time zones. Running on Microsoft Azure, Copilot uses elastic infrastructure that automatically scales GPU capacity based on demand. During traffic surges, such as simultaneous usage spikes in the US and Europe, the system rapidly adds computing resources and then scales them down during quieter periods, maintaining performance while controlling costs.

Cluster architecture is central to this capability. Large models depend on thousands of accelerators connected through high-speed interconnects such as NVIDIA NVLink or equivalent cloud technologies. These connections enable fast data transfer between machines, which is essential for both training and inference. Without efficient interconnects, performance drops and costs rise. In Copilot’s case, these clusters support both continuous model refinement and real-time inference for millions of users simultaneously.

Inference performance is especially critical in user-facing applications. Each keystroke in Copilot triggers an almost instant response. Hyperscalers achieve this through load balancing and optimised hardware, such as GPU-backed instances or specialised accelerators. Tasks are intelligently routed—simpler requests go to cost-efficient resources, while more complex queries are handled by higher-performance systems. This tiered approach ensures low latency while keeping per-query costs manageable.

Beyond compute, storage and networking are equally important. Systems like Copilot manage vast amounts of data, including model checkpoints and usage logs. Hyperscalers use tiered storage—combining durable object storage for archives with high-performance systems for active workloads. Networking infrastructure, including authentication gateways, load balancers, and regional data centres, ensures that requests are processed quickly and securely. Content delivery networks also help reduce latency and meet data residency requirements in regulated regions.

Orchestration technologies such as Kubernetes add resilience and scalability. By managing thousands of containerised services, Kubernetes enables automatic scaling, monitoring, and failover. If part of the system fails, traffic is seamlessly redirected, ensuring minimal disruption. This level of reliability is essential for tools embedded in daily workflows, where downtime directly impacts productivity and trust.

At an industry level, the scale of investment highlights the importance of this layer. By 2026, hyperscalers are expected to invest hundreds of billions of dollars into AI infrastructure, reflecting the growing demand for scalable, high-performance systems. These investments also place pressure on energy resources and supply chains, reinforcing the complexity of operating AI at scale.

For business leaders, the key insight is that generative AI strategy extends far beyond model selection. It requires designing and operating distributed systems that balance performance, cost, and reliability. Hyperscalers provide the foundation for this, enabling organisations to deliver AI services with global reach and enterprise-grade performance. Choosing the right scaling partner ensures that AI systems move from experimental tools to dependable, high-impact business solutions.

Video 6: Foundational Models Layer: LLMs and Multimodal Systems

When organisations deploy generative AI tools such as Claude, Gemini, or ChatGPT, users interact only with a polished interface. Beneath this sits the foundational models layer—the strategic core that determines capability, scalability, and adaptability. A foundation model is a large-scale pretrained system built on vast and diverse datasets to learn general-purpose representations rather than solving a single task. Unlike traditional AI systems, which were narrowly designed for specific use cases, foundation models can be adapted across multiple domains through prompting, fine-tuning, or retrieval-augmented generation, effectively turning intelligence into reusable infrastructure.

Large language models dominate this layer. Frontier systems such as GPT-5, Claude Opus 4, Gemini 2.5 Pro, and Llama 4 are trained on trillions of tokens drawn from books, code, research, and web data. Using next-token prediction, these models develop a deep understanding of language, reasoning, and conceptual relationships. This allows a single model to perform diverse tasks—summarising lengthy documents, generating code, drafting business communication, or answering complex queries. Benchmark results reflect this progress, with leading models achieving high performance across multitask knowledge, reasoning, and coding evaluations, demonstrating far greater generalisation than earlier AI systems.

The layer is now evolving into multimodal systems capable of processing text, images, audio, and video within unified architectures. Models such as Gemini 2.5 Pro and GPT-5 integrate long-context understanding—handling up to millions of tokens—alongside vision and audio capabilities. Similarly, Claude Opus 4 and Grok-4 focus on advanced reasoning while supporting multimodal inputs, and open models like Llama 3.2 extend these capabilities to more flexible deployment environments. This convergence enables systems to reason across different types of data, aligning more closely with real-world business information.

A compelling real-world application is found in healthcare through ambient AI scribes. Organisations such as Mayo Clinic and Permanente Medical Group have piloted systems like Microsoft DAX Copilot, Nabla, and Abridge. These tools transcribe consultations in real time, interpret context, structure notes, and align with clinical standards. Large-scale deployments have demonstrated measurable impact, including significant reductions in documentation time and clinician burnout. These systems go beyond transcription, combining summarisation, question answering, and structured output generation, while integrating audio input and electronic health record data under strict privacy regulations.

This example is particularly important because healthcare operates under stringent regulatory frameworks. Success in such an environment signals broader applicability across industries such as finance, insurance, and public administration, where compliance, accuracy, and data sensitivity are critical. Foundation models, however, are not standalone solutions. Enterprises enhance them using techniques like retrieval-augmented generation to ground outputs in verified data—reducing hallucinations significantly—and fine-tuning to align with domain-specific requirements.

The economic and strategic implications are substantial. The enterprise large language model market is growing rapidly, with widespread adoption across industries. At the same time, training costs for frontier models remain extremely high, often reaching tens or hundreds of millions of dollars, which

limits development to a small group of organisations such as OpenAI, Google, Anthropic, Meta, and xAI. As a result, most enterprises focus on leveraging and adapting existing models rather than building their own from scratch.

For business leaders, this represents a structural shift. The focus moves from building task-specific AI systems to leveraging reusable, adaptable engines. This accelerates development, enables faster experimentation, and simplifies integration across workflows. The key question is no longer whether AI can perform a task, but how best to adapt, govern, and integrate foundation models into business operations. Selecting models with strong capabilities, multimodal support, and a robust ecosystem—combined with techniques like retrieval-augmented generation and fine-tuning—enables organisations to achieve both flexibility and long-term competitive advantage.

Video 7: Serving Engines: vLLM & Triton

Imagine this scenario: a company launches a generative AI customer support assistant. Early feedback is positive, users appreciate the experience, and leadership expects productivity gains. Then a sudden disruption—such as a major weather event for an airline or a service outage at a bank—drives thousands of customers to the chatbot at the same time. Response times begin to slow, queues build up, and users abandon the interaction. In such situations, the issue is rarely the AI model itself; it is the serving layer—the infrastructure responsible for running the model reliably under real-world traffic.

The serving engine sits between the trained model and the application. Its role is to accept incoming requests, distribute them across compute resources, batch similar queries together, and return responses quickly. Equally important, it must do so efficiently enough to keep infrastructure costs under control. In many enterprise deployments, poor serving efficiency is one of the fastest ways AI initiatives exceed their cloud budgets, even when the underlying model performs well in controlled environments.

Consider an airline customer support assistant during severe disruptions. When thousands of passengers attempt to change flights simultaneously, query volumes can surge far beyond normal levels. The language model itself does not change, but the serving system must scale dynamically to meet demand. By batching requests and prioritising urgent cases—such as passengers already at the airport—the system can maintain acceptable response times even under heavy load, preserving both user experience and operational efficiency.

A key metric in this layer is throughput, which measures how many requests a system can process per second. In one banking workflow for automated document review, replacing a basic inference setup with an optimised serving engine increased processing capacity from around 150 documents per second to nearly 600 using the same hardware. This improvement came not from adding more GPUs, but from better batching strategies and more efficient memory management. For the business, this directly translated into faster processing and a lower cost per document.

To achieve these gains, organisations increasingly rely on specialised inference engines such as vLLM and NVIDIA Triton Inference Server. vLLM is designed specifically for serving large language models and is widely used for internal assistants and enterprise knowledge systems. Its memory-efficient architecture allows more requests to run concurrently on the same hardware, often delivering significantly higher throughput than standard deployments. NVIDIA Triton, on the other hand, supports multiple model types within a unified serving environment, enabling organisations—such as hospitals—to run imaging models alongside language models while efficiently sharing GPU resources.

For organisations deploying generative AI, the key insight is that serving infrastructure often determines whether a system succeeds in production. The same model can feel fast and cost-effective in one environment and slow and expensive in another, depending entirely on how efficiently requests are managed and resources are allocated.

In summary, while powerful models attract the most attention, it is the serving layer that ensures stability, scalability, and economic viability. When designed well, it enables AI systems to perform reliably under real-world conditions, even at peak demand.

Video 8: Quantisation & Cost Control

When organisations scale generative AI, costs can rise far faster than expected. Imagine a global retailer launching an AI product assistant across its website and mobile app. Early pilot tests appear inexpensive because only a small number of users interact with the system. But as adoption grows, daily query volumes increase rapidly. Within weeks, the company notices a sharp rise in monthly cloud costs, as every response requires GPU compute to run a large language model.

This reflects a broader industry reality. As deployments expand, inference costs often dominate overall spending, sometimes accounting for 60 to 80 per cent of total AI operating costs. Large models require significant GPU memory and compute to function. For instance, a 70-billion-parameter model such as Llama may require over 140 GB of GPU memory just to load its weights in standard 16-bit precision. At scale, serving millions of queries daily keeps GPUs running continuously, turning infrastructure into a major operational expense.

One of the most effective techniques for controlling these costs is quantisation. Quantisation reduces the numerical precision used to represent model weights and activations. Instead of using 16-bit or 32-bit floating-point numbers, models can operate with 8-bit or even 4-bit representations. This significantly lowers memory requirements and computational demand, enabling more efficient use of hardware without fundamentally changing the model's structure.

In practice, the performance trade-off is often minimal. Many benchmarks show that 8-bit quantisation typically reduces accuracy by less than one to two per cent on tasks such as summarisation or question answering. Even 4-bit quantisation may result in only modest performance loss, particularly when supported by advanced optimisation techniques. This makes quantisation a practical choice for high-volume enterprise applications where efficiency matters as much as precision.

A real-world example comes from a European regulatory technology firm that develops AI tools to summarise complex financial compliance documents. These documents can span hundreds of pages, requiring substantial compute per request. By shifting to 8-bit quantisation, the company was able to process roughly twice as many summaries per GPU hour. This delivered significant cost savings while maintaining output quality that analysts could rely on.

Similar patterns appear across industries. Healthcare organisations use quantised models to generate clinical notes and summarise patient records on secure internal systems, reducing hardware requirements while preserving data privacy. Telecommunications companies apply quantisation in customer support chatbots, where millions of interactions occur each month. In many cases, high-volume queries are handled by lower-precision models, while more complex requests are routed to higher-precision systems for deeper reasoning.

The key insight is that quantisation is not merely a technical optimisation; it is a strategic cost control mechanism. By reducing memory usage and increasing throughput, organisations can handle

significantly higher workloads using the same infrastructure. When applied effectively, it enables scalable, efficient, and economically sustainable generative AI deployments.

Video 9: Final Thoughts

As we conclude this week, it is important to recognise that generative AI may appear as a simple application interface, but what truly powers it is a deeply engineered stack. We began by examining this layered architecture—from silicon and specialised compute, through cloud infrastructure, foundational models, serving engines, and finally applications. Each layer contributes distinct performance, cost, and risk characteristics that ultimately shape enterprise outcomes.

We then explored how constraints emerge across the stack. Compute limitations directly influence latency, while energy availability determines how far systems can scale. Governance challenges arise from data handling, privacy requirements, and deployment decisions. Hardware choices—whether general-purpose GPUs or specialised accelerators—affect memory bandwidth, efficiency, and the balance between training and inference. These are not abstract technical considerations; they have direct implications for business performance and sustainability.

At the infrastructure level, hyperscalers such as Microsoft Azure, Amazon Web Services, and Google Cloud Platform enable scale through distributed storage, networking, and orchestration. Above this, foundation models provide generalisable intelligence across text, image, and audio tasks. Serving engines such as vLLM and NVIDIA Triton Inference Server transform static models into responsive systems optimised for throughput and reliability. Techniques such as quantisation further ensure that deployments remain economically viable by balancing accuracy with cost efficiency.

Taken together, these layers reveal an essential truth: generative AI is not merely software—it is engineered infrastructure with economic, geopolitical, and operational implications. Leaders who understand this stack from end to end are far better equipped to evaluate vendors, control costs, manage risk, and scale responsibly.

Ultimately, strategic advantage will belong to organisations that treat generative AI not as a standalone tool, but as a full-stack capability—one that integrates technology, operations, and governance into a cohesive, scalable system.